

SUWON AB, Korea, Republic of

WMO: 471200

Lat: 37.239N

Lon: 127.007E

Elev: 27

StdP: 101.00

Time Zone: 9.00 (E09)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.8	-11.8	-18.7	0.7	-10.8	-16.2	0.9	-9.6	10.6	-3.2	9.2	-3.6	0.5	300	0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.1	33.1	26.0	31.9	25.1	30.2	24.2	26.7	31.3	26.2	30.3	25.5	29.0	2.9	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	29.4	25.0	20.2	28.4	24.2	19.2	27.4	84.2	31.7	81.4	30.2	78.4	28.5	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.0	5.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			-15.4	28.0	3.0	0.8	-17.6	28.6	-19.4	29.1	-21.1	29.6	-23.3	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-3.6	-0.7	5.5	12.1	17.7	22.3	25.7	26.3	21.0	14.4	7.4	0.7
	DBStd	10.61	4.00	4.52	3.26	3.77	2.60	2.26	2.25	2.32	2.93	3.72	4.09	4.09
	HDD10.0	1278	421	301	143	20	0	0	0	0	10	97	287	
	HDD18.3	2857	679	534	399	191	43	2	0	9	127	328	546	
	CDD10.0	2183	0	0	2	82	240	371	486	507	329	147	19	0
	CDD18.3	718	0	0	0	3	24	123	228	248	88	5	0	0
	CDH23.3	5623	0	0	0	29	186	774	1840	2194	555	43	0	0
	CDH26.7	1791	0	0	0	2	23	202	616	830	117	2	0	0
Wind	WSAvg	1.9	1.9	2.1	2.3	2.3	2.0	1.7	1.8	1.7	1.5	1.6	1.7	1.7
Precipitation	PrecAvg	1296	25	30	47	92	87	119	336	280	148	55	54	23
	PrecMax	2044	90	116	127	396	245	414	568	851	583	264	151	54
	PrecMin	850	2	1	5	7	12	5	145	53	3	0	9	1
	PrecStd	272	22	25	29	86	53	103	140	156	119	61	35	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	12.9	18.2	26.1	29.0	32.0	35.0	35.0	31.2	26.2	19.9	13.4
		MCWB	4.0	7.1	10.7	16.6	19.2	22.3	26.5	26.6	23.8	18.5	14.7	9.0
	2%	DB	5.9	10.2	15.8	23.9	27.1	30.2	33.0	33.8	29.8	24.2	17.8	10.8
		MCWB	3.0	6.3	9.0	16.1	18.3	21.6	26.2	26.4	23.0	17.6	12.8	7.0
	5%	DB	4.2	8.2	13.8	21.8	25.5	29.0	31.2	32.2	28.0	22.9	16.0	9.0
		MCWB	2.0	5.2	8.3	14.1	17.4	21.1	25.2	25.8	21.9	16.7	11.5	6.4
	10%	DB	2.9	6.2	11.8	19.4	23.9	27.6	30.0	31.0	26.3	20.9	14.1	7.1
		MCWB	1.0	3.4	7.1	13.0	16.5	20.7	24.7	25.3	20.9	15.8	10.7	4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	9.2	12.1	19.4	21.7	24.5	27.6	27.9	25.3	20.1	15.7	9.9
		MCDB	6.7	11.3	15.7	23.0	25.8	29.1	32.4	33.1	28.7	23.7	18.6	11.8
	2%	WB	3.5	6.8	10.0	16.8	20.0	23.2	26.7	27.0	24.3	18.9	14.0	7.8
		MCDB	5.2	8.9	14.5	22.0	24.4	27.3	31.2	31.7	27.5	22.6	16.7	9.6
	5%	WB	2.3	5.2	8.6	15.2	18.8	22.4	26.1	26.5	23.3	17.6	12.5	6.4
		MCDB	4.2	8.3	12.5	20.1	23.3	26.6	30.0	30.8	26.4	21.4	14.9	8.5
	10%	WB	0.9	3.7	7.4	13.7	17.7	21.6	25.5	25.9	22.2	16.5	11.1	4.9
		MCDB	2.4	6.1	11.1	18.4	22.0	25.7	28.8	29.8	25.5	19.9	13.6	7.0
Mean Daily Temperature Range	5% DB	MDBR	9.9	9.9	10.7	12.4	11.1	9.6	7.3	8.1	10.1	11.6	10.7	9.6
		MCDBR	10.0	11.4	13.9	15.2	13.6	12.0	9.5	9.9	10.6	13.4	12.6	11.3
		MCWBR	7.8	7.9	8.5	8.2	6.4	5.3	4.1	4.2	5.0	7.4	8.3	8.4
	5% WB	MCDBR	9.7	10.2	12.5	12.6	11.0	9.6	8.0	8.9	8.6	11.5	10.6	10.3
		MCWBR	7.9	7.2	8.4	7.6	6.2	4.9	3.9	4.1	4.8	7.2	8.0	8.3
Clear-Sky Solar Irradiance	taub	0.440	0.490	0.573	0.613	0.626	0.698	0.662	0.582	0.510	0.499	0.482	0.437	
	taud	1.947	1.858	1.714	1.675	1.696	1.618	1.740	1.928	2.053	1.958	1.944	1.973	
	Ebn at Noon	695	712	693	697	698	648	668	714	744	697	647	664	
	Edh at Noon	140	173	219	240	239	259	227	185	154	154	140	128	
All-Sky Solar Radiation	RadAvg	2.48	3.26	4.28	4.98	5.59	5.20	4.12	4.36	4.12	3.59	2.45	2.18	
	RadStd	0.15	0.26	0.43	0.36	0.52	0.52	0.62	0.62	0.45	0.26	0.25	0.15	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page